

Dayton, OH
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Matthew Dies

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Lead Software Engineer and Supervisor with 5+ years of experience advancing software through R&D, full-stack application development, and production infrastructure. Proven track record of architecting mission-critical platforms, including spearheading the ground-up microservices re-architecture of a flagship knowledge graph application and standardizing developer ecosystems company-wide. Expert in Python, containerization, and automated CI/CD pipelines, with a focus on delivering scalable, cloud-native and on-premises solutions that transform experimental concepts into high-performance, enterprise-ready systems.

Work Experience

Lead Software Engineer, Supervisor Platform Team	Tenet3 Dayton, OH	2026 - Present
- Spearheaded the ground-up backend re-architecture of MeTRA®, the company's flagship knowledge graph modeling web application, building out a FastAPI microservices architecture. - Designed and developed a core internal Python package establishing universal system patterns, including transaction integrity, standardized NATS streaming configurations, file storage, and more. - Architected the project's deployment infrastructure by overseeing modular Helm charts and partnering with IT/Ops to streamline production delivery. - Built foundational GitLab CI Components library, devcontainers, and project templates, standardizing the development lifecycle across the company.		
Lead Software Engineer Advanced Capabilities Team	Tenet3 Dayton, OH	2025 - 2026
- Created a "microservice deployer" framework acting as the company's primary way to rapidly advance R&D technologies including document parsing and summarization, data extraction, and automatic knowledge model creation into production. - Mentored interns and junior software engineers, implementing growth paths that successfully converted high-performers into full-time engineering roles.		
Software Engineer Advanced Capabilities Team	Tenet3 Dayton, OH	2022 - 2025
- Engineered a standardized devcontainer ecosystem (Python/Docker) used as the primary development environment for 50%+ of the company's technical staff, spanning two teams. This includes a Python package for container configuration file management + build automation, curated Docker images, pre-built CI templates, remote development capability with automatic GPU integration, and version-controlled documentation. - Developed a binary data parser that reduced processing time from hours to seconds. - Deployed a website that parsed unstructured data into a database and displayed the information in a dashboard, reducing man hours by over 600%.		
Data Scientist Research and Development Team	Tenet3 Dayton, OH	2021 - 2022
- Implement large-scale cloud computational algorithms for exact subgraph matching on various Amazon Web Services including S3, EC2, EMR, Lambda, and CloudWatch. - Created Python scripts to fetch known vulnerabilities for software and hardware within knowledge models and supplement this information with vulnerability metrics and weaknesses from using the NVD, CVE, and CWE APIs.		

Technologies and Languages

- **Languages:** Python, SQL (PostgreSQL), Bash, HTML/CSS (Tailwind CSS).
- **Frameworks, Libraries & Tools:** FastAPI, SQLAlchemy/SQLModel, Django, Flask, Pydantic, Pytest, uv, ty, Ruff, Poetry.
- **Infrastructure & Tools:** Docker, Podman, Devcontainers, GitLab CI/CD.
- **Cloud (AWS):** ECS, Lambda, RDS, S3, CDK.
- **Other:** Technical Leadership, Mentorship, RESTful API Design, Microservices Architecture.

Education

- **B.S. Applied Mathematical Economics**, University of Dayton, OH. **2017 - 2020**
Details: 3.83 GPA, Magna Cum Laude, mathematical honors society Pi Mu Epsilon member

Projects

- **Home Server:** I maintain a home server on my headless Ubuntu PC where I run and maintain containers, automation scripts, and more. Highlights include using Caddy as a reverse proxy combined with Authelia for authentication, hosting numerous webpages including self-made containers. I often joke that the server is so automated nowadays that I've "hobbiied myself out of a hobby!" Check it out on [GitHub](#).
- **Dashboard for my Family's Football Pool:** Built using Flask and TailwindCSS, productionized using Docker, and deployed/self-hosted using Docker Compose, Caddy, and CloudFlare. Uses Pydantic to validate live results fetched from ESPN's API to display scores of games from the current week, and uses Postgres to store the assignments and winners for the pool! Given a chance to do a refactor, I'd love to replace my usage of `requests` with `niquests` or `httpx2` for HTTP/2+ protocol usage. Check it out live at footballpool.mattdies.com, or on [GitHub](#).
- **Self-hosted LLM:** My home server has some GPUs sitting on it already, so I've spun up a few LLMs that myself and my fiancée query with [Open WebUI](#). It's surprisingly powerful and easy to use, while being great for privacy.
- **In progress...** While I'm not planning for my wedding and cooking up new meals, I'm tinkering around with a few new projects; a (password-protected) webapp to store funny quotes from my friends, and a pantry inventory tracking app that I want to eventually integrate directly with Mealie.

Interests

- I fell in love with cooking after self-hosting [Mealie](#) and have been working to perfect my risotto recipe. While I've traditionally loved cooking meals for dinner, I've recently been expanding my skillset into baking bread and the occasional desserts as well!
- I play in an adult baseball league as a catcher and am a huge fan of the Cleveland Guardians.
- I've completed two half-marathons (or one *incredibly* slow full marathon), and am training for a third. I've also recently started lifting weights, and have discovered that it's not as intimidating as it seems. Maybe one of these days I'll get one of those cool "muscles" that everyone is talking about.